

Design Pattern 6: Proxy Pattern

Exercise 6: Implementing the Proxy Pattern

Scenario:

You are developing an image viewer application that loads images from a remote server. Use the Proxy Pattern to add lazy initialization and caching.

Steps:

Create a New Java Project:

Create a new Java project named ProxyPatternExample.

Define Subject Interface:

Create an interface Image with a method display().

Implement Real Subject Class:

Create a class RealImage that implements Image and loads an image from a remote server.

Implement Proxy Class:

Create a class ProxyImage that implements Image and holds a reference to RealImage.

Implement lazy initialization and caching in ProxyImage.

Test the Proxy Implementation:

Create a test class to demonstrate the use of ProxyImage to load and display images.